other. The net force is zero in y direction. So in y-direction there is no acceleration. The forces F_3 and F_4 are not equal in magnitude and F_3 is greater than F_4. So there is a net acceleration in negative x direction

Case (d):

The forces F_1 and F_2 are equal in magnitude and act opposite to each other. The net force is zero in y direction. So there is no acceleration in y-direction. The forces F_3 and F_4 are not equal in magnitude. The force F_4 is greater than the force F_3. So there is a net acceleration in the positive x direction.

3. A gun weighing 25 kg fires a bullet weighing 30 g with the speed of 200 ms^{-1}. What is the speed of recoil of the gun.

Solution:

Mass of the gun M = 25 kg
Mass of the bullet m = 30 g = 30×10^{-3} kg
Speed of bullet v = 200 m s^{-1}
Speed of gun V = ?
The motion is in one dimension.
As per law of conservation of momentum,

$$MV + mv = 0$$

$$V = \frac{-mv}{M}$$

$$V = \frac{-30 \times 10^{-3} \times 200}{25} = -240 \times 10^{-3} m s^{-1}$$

The negative sign shows that the gun moves in the opposite direction of the bullet. Further the magnitude of the recoil speed is very small compared to the bullet's speed.

4. A wooden box is lying on an inclined plane. What is the coefficient of friction if the box starts sliding when the angle of inclination is 45°.

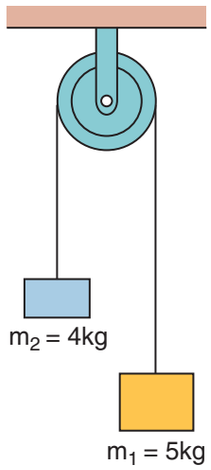
Solution:

Angle of inclination $\Theta = 45°$
$\therefore$ Coefficient of friction $\mu = \tan \Theta = \tan 45° = 1$

5. Two masses m_1 = 5 kg and m_2 = 4 kg tied to a string are hanging over a light frictionless pulley. What is the acceleration of each mass when left free to move? (g = 10 m s^{-2})

$$a = \frac{m_1 - m_2}{m_1 + m_2} \times g$$

$$= \frac{5-4}{5+4} \times 10 = \frac{1}{9} \times 10 = 1.1\ m s^{-2}$$



6. A block of mass m is pushed momentarily along a horizontal surface with an initial velocity u. If μ_k is the coefficient of kinetic friction between the object and surface, find the time at which the block comes to rest.

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